

Experimental Plan: Curvature-Aware Step-Size Adaptation and Guidance Damping for Optimal Transport Flow Matching

Generative Learning and Diffusion Models Project

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1 Overview & Objectives

The primary objective of this experiment is to validate whether local vector field acceleration $a_t = \frac{dv_t}{dt}$ can serve as a computationally lightweight, training-free diagnostic signal to co-adapt ODE integration step sizes and Classifier-Free Guidance (CFG) scales in Optimal Transport Conditional Flow Matching (OT-CFM). The design prioritizes low computational footprint, enabling full training and inference evaluation within 90 minutes on a standard consumer GPU.

2 Datasets & Architecture Specifications

- **2D Synthetic Distributions (Swiss Roll and Pinwheel):**

- *Architecture:* 3-layer Multi-Layer Perceptron (MLP) with 128 hidden units per layer, SiLU non-linearities, and sinusoidal time embeddings.
- *Training:* 5,000 steps using Adam optimizer ($\text{lr} = 10^{-3}$, batch size 512).

- **Image Benchmark (Fashion-MNIST):**

- *Resolution & Preprocessing:* 28×28 grayscale images zero-padded to 32×32 , normalized to $[-1, 1]$.
- *Architecture:* Lightweight score/velocity U-Net ($< 2\text{M}$ parameters) featuring 3 down-sampling/upsampling blocks and conditional class embeddings.
- *Guidance Training:* 10% unconditional dropout ($c = \emptyset$) applied during training for Classifier-Free Guidance (CFG).
- *Training:* 15 epochs ($\approx 14,000$ steps) with AdamW ($\text{lr} = 2 \times 10^{-4}$, batch size 128).

- **Training Objective:** Both models are optimized using the standard OT-CFM regression objective:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0,1], x_0 \sim p_0, x_1 \sim p_1} [\|v_\theta(x_t, t, c) - (x_1 - x_0)\|_2^2], \quad (1)$$

where $x_t = (1 - t)x_0 + tx_1$ defines the linear conditional path.

3 Solver Implementation & Adaptation Mechanisms

To evaluate trajectories without incurring backpropagation passes, the algorithm caches intermediate evaluations using a modified Heun-type predictor-corrector framework:

1. **Guidance Vector Field:** At time t , given base guidance scale w , the guided velocity field is:

$$v_{\text{guided}}(x_t, t, c) = v_{\theta}(x_t, t, \emptyset) + w_{\text{eff}}(t) \cdot (v_{\theta}(x_t, t, c) - v_{\theta}(x_t, t, \emptyset)). \quad (2)$$

2. **Acceleration Caching:** Using trial step Δt , the predictor stage computes $v_1 = v_{\text{guided}}(x_t, t, c)$ and trial state $\tilde{x} = x_t + \Delta t \cdot v_1$. The corrector stage evaluates $v_2 = v_{\text{guided}}(\tilde{x}, t + \Delta t, c)$. Acceleration is computed via finite differences:

$$a_t \approx \frac{v_2 - v_1}{\Delta t}. \quad (3)$$

3. **Step-Size Adaptation:** The step size is updated dynamically according to local trajectory curvature:

$$\Delta t_k = \min \left(\Delta t_{\text{max}}, \frac{\eta}{\|a_t\|_2 + \epsilon} \right), \quad (4)$$

where $\epsilon = 10^{-5}$ prevents division by zero, and $\eta \in \{0.05, 0.1\}$.

4. **Curvature-Damped Guidance:** To prevent extreme extrapolation off the data manifold, the effective guidance weight is modulated as:

$$w_{\text{eff}}(t) = \frac{w}{1 + \gamma \|a_t\|_2}, \quad (5)$$

with damping hyperparameter $\gamma \in \{0.1, 0.5\}$.

4 Ablation Matrix & Experimental Configurations

The experimental validation is structured as a 2×2 factorial design across discrete sampling budgets $N \in \{4, 8, 12, 16\}$ and nominal guidance weights $w \in \{1.5, 3.0, 5.0, 7.0\}$:

Configuration	Step Sizing	CFG Schedule	Primary Hypothesis Tested
M1 (Baseline Euler/Heun)	Uniform ($\Delta t = 1/N$)	Fixed $w_{\text{eff}} = w$	Standard baseline performance
M2 (Step-Adaptive Only)	Dynamic $\Delta t_k(a_t)$	Fixed $w_{\text{eff}} = w$	Truncation error reduction
M3 (Damping-Only)	Uniform ($\Delta t = 1/N$)	Damped $w_{\text{eff}}(a_t)$	Off-manifold artifact suppression
M4 (Full Proposed Method)	Dynamic $\Delta t_k(a_t)$	Damped $w_{\text{eff}}(a_t)$	Joint trajectory stabilization

Table 1: Ablation configurations evaluated across all datasets and budgets.

5 Quantitative Metrics & Visualization Deliverables

• Synthetic Evaluation (2D):

- Compute Wasserstein-2 (W_2) distance between 2,000 generated samples and ground-truth test points.
- Generate phase-portrait scatter plots visualizing individual trajectory curves under high guidance ($w = 5.0$) comparing M1 vs. M4.
- Plot $\|a_t\|_2$ profiles across normalized time $t \in [0, 1]$ to demonstrate boundary curvature spikes.

• Image Benchmark Evaluation (Fashion-MNIST):

- Compute Fréchet Inception Distance (FID) on 10,000 generated samples against the validation split.
- Report effective Number of Function Evaluations (NFEs) per sample to establish Pareto frontiers (FID vs. NFE).